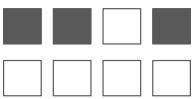
a

$$Y_{g,s,b,c} = \mu_g + k\beta_g C_s + u_{g,s} + b_{g,b} + \varepsilon_{g,s,b,c}$$

Protein mean Condition effect Sample effect Batch effect Cell noise

$$u_{g,s} \sim N(0, \sigma^2_{s,g}) \; ; \; b_{g,b} \sim N(0, \sigma^2_{b,g}) \; ; \; \varepsilon_{g,s,b,c} \sim N(0, \sigma^2_{e,g})$$

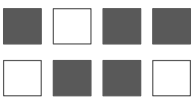
Carrier TMT: identify plex, then quantify channels



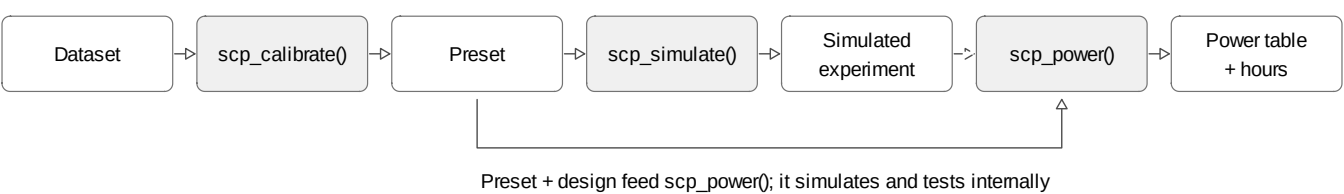
$$P(I_{g,b} = 1) = p_g$$

$$P(Q_{g,b,c} = 1 \mid I_{g,b} = 1) = q_g$$

DIA / label-free: quantify each cell



$$\text{logit}(P(M_{g,c} = 1)) = \text{logit}(r_g) + \lambda(x - \mu_g)$$

b

SCP measurements + bulk donor variance